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Interpretable Federated Learning for Fraud Detection in Non-IID Blockchain Transactions

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Highlights

- Proposes a novel Federated Learning framework for fraud detection in blockchain transactions that preserves client privacy under IID and non-IID settings.

- Demonstrates that Tabular Transformer with Adam optimizer achieves over 99offers robust performance under high data heterogeneity.

- Integrates SHAP and LIME to provide both global and local explainability, identifying token-specific and temporal features as key fraud indicators.

- Offers practical guidance on model and optimizer selection based on data distribution and computational constraints in decentralized environments.

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Interpretable Federated Learning for Fraud Detection in Non-IID Blockchain Transactions

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ABSTRACT

Detecting fraud in decentralized blockchain transactions presents significant challenges due to privacy constraints, scalability issues, and the need for interpretability in decision-making. Traditional machine learning and deep learning approaches rely on centralized data aggregation, which compromises privacy and introduces single points of failure. Federated Learning (FL) offers a decentralized, privacy-preserving alternative but struggles with highly non-IID data distributions and lacks built-in interpretability. To address these limitations, this study introduces a novel framework that integrates FL with Explainable AI (XAI) to enable secure, scalable, and transparent fraud detection in blockchain transactions. Our contributions include the development of federated optimization strategies specifically designed for non-IID environments, ensuring fairness and efficiency across diverse client nodes. Experimental results demonstrate that the Tabular Transformer model with the Adam optimizer achieves over 99% accuracy, precision, recall, and F1 score under IID settings. In contrast, a 2-layer Neural Network model trained with SGD shows superior robustness under non-IID conditions, achieving 96.24% to 98.98% accuracy across both partial and full client participation with minimal computational cost. XAI techniques such as SHAP and LIME further enhance the system's transparency by identifying token-specific statistical features (e.g., maximum, minimum, average, and uniqueness of 'ERC20' tokens) as primary contributors to model decisions. This work highlights the promise of combining FL and XAI for developing privacy-preserving and interpretable fraud detection systems and sets the foundation for future research on lightweight, ethical AI solutions in decentralized, resource-constrained environments.

1. Introduction

Blockchain, a decentralized and distributed ledger technology, has revolutionized digital financial systems by offering transparency, authenticity, immutability, and security in transaction recording [1, 2]. Transactions are grouped into cryptographically linked blocks, forming a secure chain that eliminates the need for centralized oversight [3]. This decentralized architecture is a key enabler for applications such as cryptocurrency, supply chain management, and identity verification. However, like any complex system, blockchain also has its vulnerabilities. Its anonymous and decentralized nature makes it susceptible to fraudulent activities such as double-spending, phishing, and Sybil attacks. These threats undermine user trust and challenge the security of blockchain ecosystems. As such, there is an urgent need for effective, scalable, and privacy-preserving fraud detection mechanisms in blockchain transactions.

The unique characteristics of blockchain, its distributed structure and peer-to-peer transaction mechanism present both opportunities and challenges for fraud detection. While blockchain eliminates intermediaries, enhancing performance and reducing costs, it also complicates the detection of fraudulent behavior. Traditional fraud detection methods rely on centralized data access and analysis, which conflicts with the distributed nature of blockchain systems [4, 5]. Additionally, the massive volume of transactions, the diversity of user behavior, and the need for real-time detection significantly complicate fraud identification efforts [6, 7]. Since fraudulent activities often mimic legitimate behavior, advanced analytics are required to detect subtle anomalies across large-scale, high-dimensional data. Machine Learning (ML) and Deep Learning (DL) offer promising solutions by learning complex patterns indicative of fraud. However, their dependence on centralized data aggregation introduces privacy concerns and conflicts with the blockchain's

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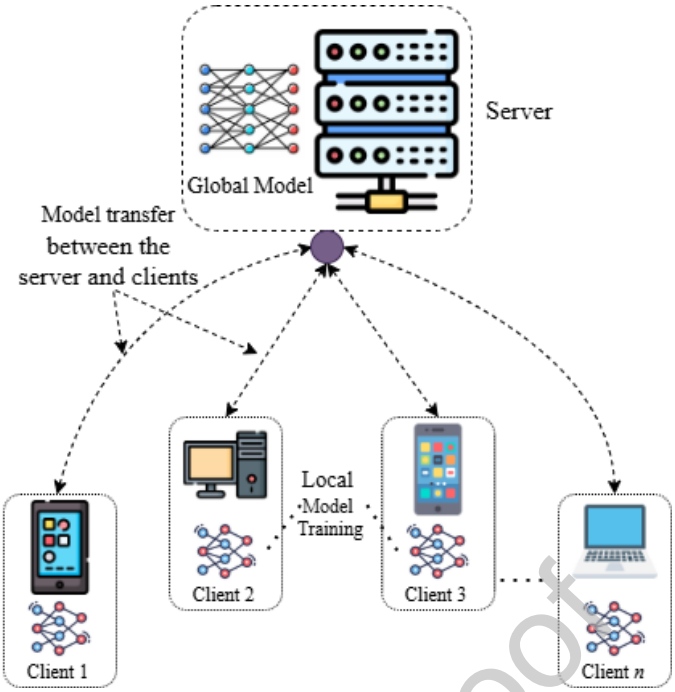


Figure 1: The working process of FL. The server distributes the global model to multiple clients, which perform local training using their private data. The updated local models are then sent back to the server for aggregation to improve the global model. This process repeats over multiple rounds while keeping client data local to ensure privacy.

decentralized architecture [8, 9]. Furthermore, these models often struggle with data heterogeneity, scalability, and the need for real-time responses in dynamic environments.

Federated Learning (FL) has emerged as a promising alternative that aligns with blockchain's decentralized principles. FL enables multiple devices or nodes to collaboratively train ML models without sharing raw data [10, 11, 12, 13]. In the context of blockchain, each participant (e.g., wallets or exchanges) uses local transaction data to train a model and shares only model updates with a central aggregator [14, 15, 16]. These updates are combined to form a global model, refined iteratively over multiple communication rounds. This paradigm preserves data privacy, complies with data protection regulations, and enhances user trust, making FL a suitable choice for decentralized fraud detection [17, 18]. Figure 1 illustrates the typical workflow of an FL system. Its compatibility with blockchain's distributed structure enables collaborative model development without compromising user data, thereby facilitating more secure and scalable fraud detection solutions.

Despite its advantages, FL introduces challenges in interpretability. In high-stakes domains like financial fraud detection, it is essential not only to make accurate predictions but also to understand the rationale behind those decisions. Explainable AI (XAI) addresses this gap by providing insight into model behavior and highlighting the key features that drive predictions [19]. By integrating XAI techniques with FL, stakeholders can gain both global and local interpretability of the models, increasing transparency, fostering trust, and supporting compliance with regulatory standards. For instance, XAI enables users to see which transaction features triggered a fraud alert, providing justifications that align with human reasoning. This combination of privacy preservation and interpretability provides a powerful solution for secure and transparent fraud detection in blockchain systems [20].

Recent studies have highlighted limitations in traditional and centralized fraud detection frameworks. These include limited robustness to heterogeneous data and inadequate scalability in distributed environments with non-Independent and Identically Distributed (non-IID) characteristics [21]. In blockchain networks, clients such as exchanges and wallets often exhibit vastly different transaction patterns and fraud profiles, leading to biased local updates that degrade global model performance [22]. The absence of representative data across clients further hinders generalization. In addition, the "black-box" nature of many ML models limits their interpretability, impeding stakeholder trust and regulatory adoption [23]. FL helps mitigate these issues by enabling decentralized model training, but it introduces communication overhead, inconsistent performance between clients, and lacks built-in interpretability features. Meanwhile, XAI offers

transparent explanations but has yet to be widely integrated into FL frameworks in the context of blockchain fraud detection [24].

To address these challenges, this study proposes a hybrid framework that combines FL and XAI for privacy-preserving, scalable, and interpretable fraud detection in blockchain transactions. FL eliminates the need for centralized data aggregation, supports distributed collaboration, and improves detection performance. XAI adds critical transparency by identifying influential features and explaining individual predictions. Together, they offer a unified solution that meets the practical demands of real-world blockchain systems, especially in security-sensitive applications [25]. The primary objectives of this research are to enhance the reliability, scalability, and interpretability of FL-based fraud detection systems across diverse data distributions. We investigate the performance of various model-optimizer configurations in both IID and non-IID settings to identify optimal trade-offs between accuracy and computational efficiency. By integrating XAI, we provide stakeholders with clear explanations of model predictions at both global and local levels. The proposed approach addresses existing limitations by introducing adaptive strategies that preserve privacy, ensure fairness, and enable trust in AI-powered fraud detection. This work lays the foundation for future advancements in decentralized and ethical AI systems. The key contributions of this research are as follows:

- We develop robust FL optimization strategies tailored to both IID and various levels of non-IID data distributions for fraud detection in blockchain transactions, enhancing scalability, fairness, and performance in decentralized environments.
- We integrate XAI methods to improve the interpretability of model predictions by identifying and visualizing key features using global and local explanation techniques such as Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME).
- We present a unified FL-XAI framework that enables privacy-preserving, scalable, and reliable fraud detection in decentralized blockchain transactions.
- We address ethical concerns in AI-driven decision-making by ensuring transparency, accountability, and fairness while maintaining compliance with privacy regulations through both simple and complex FL architectures.

The remainder of this paper is organized as follows: Section 2 reviews related work in blockchain fraud detection and FL-based approaches. Section 3 details the proposed system and methodology. Section 4 presents experimental results and analysis. Finally, Section 5 concludes the paper and outlines future research directions.

2. Related Work

Fraud detection in blockchain transactions has garnered increasing research interest due to the widespread adoption of blockchain in various sectors, including finance, healthcare, and supply chain systems [26, 27]. This growing significance has encouraged the integration of blockchain analytics with ML techniques to develop advanced, data-driven fraud detection mechanisms [28]. This section reviews recent studies focused on blockchain security, transaction integrity, and decentralized fraud detection methods within blockchain transactions. In the financial domain, the adoption of blockchain has transformed digital transaction processes but also introduced new vulnerabilities, leading to a surge in cybercrime targeting blockchain-based financial systems. For example, Ethereum's pseudonymous nature has been linked to scams and money laundering activities, drawing considerable research attention [29]. In response, various studies have explored the fusion of blockchain and ML to develop robust fraud detection systems. One such study proposed a framework that combines logistic regression (LR), support vector machines (SVM), adaptive boosting (AdaBoost), and ensemble stacking to identify fraudulent behavior on the Ethereum network [30, 31]. Among these, the ensemble methods achieved the highest performance, with an F1 score of 99.6%, due to their ability to capture transaction behavior through advanced feature extraction.

Other optimized ML models have also demonstrated strong performance. A Light Gradient Boosting Machine (LGBM) model achieved 99.03% accuracy in Ethereum fraud detection, outperforming XGBoost and Random Forest (RF) after hyperparameter tuning [32]. Similarly, the Eth-PSD framework effectively detected Ethereum phishing scams by addressing class imbalance and feature complexity, achieving 98.11% accuracy with a false positive rate of just 0.01% [33]. A separate study integrated blockchain's secure data storage with ML classifiers, where RF achieved 97.5% accuracy in cryptocurrency fraud detection, demonstrating the practical potential of combining secure

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infrastructure with intelligent analytics [34]. They use LR, RF, KNN, AdaBoost, and XGBoost where XGBoost emerged as the most effective in recognizing fraudulent transaction patterns.

Ensemble methods continue to show promise. In [35], CatBoost and LGBM models achieved 97%–98.42% accuracy when combined with K-Means-SMOTE for class balancing. This study reinforced the effectiveness of ensemble learning, particularly when addressing class imbalance in cryptocurrency systems. On the other hand, fraud detection in smart contracts remains a challenge due to their complex inter-node interactions. Traditional ML models like One-Class SVM and Isolation Forest struggle in such scenarios. A recent solution involved One-Class Graph Neural Networks (GNNs), which outperformed non-graph-based methods in detecting anomalies in blockchain transactions [36]. Further, heterogeneous GNNs, such as Relational Graph Convolutional Networks (RGCN), demonstrated better performance than homogeneous models in detecting phishing and malicious events across blockchain transactions, underscoring the importance of model architecture and hyperparameter tuning [37].

Model interpretability is another critical area. The study in [19] tackled the "black box" problem by combining XAI techniques with tree-based ensemble models. The proposed XGBCLUS under-sampling algorithm enhanced both True Positive Rate (TPR) and ROC-AUC scores, making the detection process more transparent and robust. Similarly, bot detection on blockchain platforms has been addressed using supervised and unsupervised learning, with Gaussian Mixture Models and RF classifiers yielding high accuracy in identifying automated trading behaviors and market manipulation tactics [38]. Another recent contribution, BERT4ETH, applied Transformer-based architectures to capture sequential patterns in high-frequency Ethereum transactions. This approach improved phishing detection and de-anonymization performance over conventional graph-based models [39].

The rapid growth of online transactions on Ethereum has also led to the emergence of new fraud types. In response, a DL model integrated with a hybrid Genetic Algorithm namely Cuckoo Search (GA–CS) optimization strategy was proposed, outperforming conventional classifiers such as KNN, LR, MLP, XGBoost, LGBM, RF, and SVC in fraudulent transaction detection [40]. Another critical area of concern is Ponzi schemes, which exploit the blockchain's anonymity. A study in [41] introduced a data mining-based detection model, leveraging a custom-built dataset of legitimate and Ponzi smart contracts. This model provided accurate and reliable Ponzi scheme detection, potentially usable at the smart contract design stage. Finally, the convergence of cryptocurrencies and traditional finance systems has opened new avenues for illicit activities such as money laundering and terrorism financing, further emphasizing the need for intelligent, adaptive, and interpretable fraud detection frameworks in blockchain ecosystems [42]. Additionally, another study [43] proposed a blockchain-enabled semi-distributed learning framework for secure and low-latency computation offloading in space–air–ground integrated power IoT systems. Their approach combines federated deep reinforcement learning with resource optimization to minimize queuing delay while maintaining long-term security constraints, demonstrating improved performance in delay reduction and secure distributed computation.

3. Methodology

This section presents the proposed privacy-preserving and explainable fraud detection framework for decentralized blockchain transactions. We begin with an overview of the approach and its key components, followed by detailed descriptions of the dataset, data distribution strategies, FL setup, deep learning models and optimizers, and the applied XAI techniques.

3.1. Approach Overview

The proposed methodology begins with data preprocessing to ensure the dataset is suitable for training ML models. Missing values are handled appropriately, and irrelevant or redundant features are removed. The processed dataset is then partitioned into IID and non-IID distributions across multiple clients, enabling decentralized fraud detection through FL. Two model architectures are considered: a Simple Neural Network (2-layer) and a Tabular Transformer. Each model is trained under different optimizer configurations (SGD, Adam, and Adagrad) to evaluate their performance under varying data heterogeneity levels. Federated Averaging (FedAvg) is employed to aggregate local model updates and produce a global model. Finally, explainability is achieved by integrating XAI methods, including Shapash (based on SHAP) and LIME, which provide both global and local explanations. Fig. 2 illustrates the overall framework.

3.2. Dataset and Data Preprocessing

We utilize the publicly available Ethereum Fraud Detection Dataset from Kaggle [44], which comprises transactional data and metadata for both legitimate and fraudulent Ethereum transactions. Key features include transaction

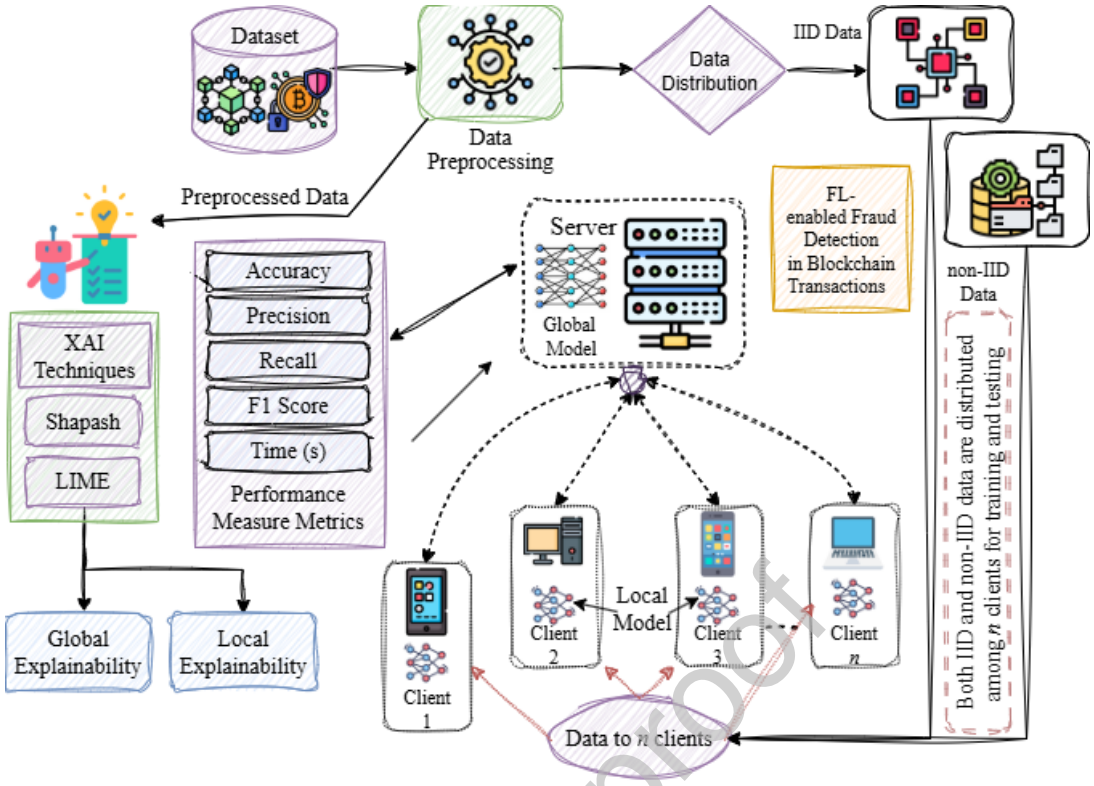


Figure 2: Overview of the proposed blockchain transaction fraud detection system using FL and XAI. The dataset is first preprocessed and distributed among multiple clients under both IID and non-IID settings. Each client performs local model training, and the updates are sent to the server to build a global model through aggregation. The framework is evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, and execution time. Additionally, XAI techniques such as Shapash and LIME are applied to provide both global and local explainability of the model decisions.

frequency, Ether value metrics (min, max, average), account addresses, contract interactions, and token-specific details (e.g., ERC20 usage statistics). These features provide a comprehensive basis for detecting fraudulent behavior. Missing values in continuous features are imputed with the mean, while categorical features are filled using the mode. Columns with all-zero values or irrelevant information are removed to streamline the dataset and enhance its compatibility with ML models.

3.3. Data Distribution Strategies

To evaluate the framework's robustness, we simulate both IID and non-IID data distributions across 20 clients. For the IID setting, each client receives an equal number of randomly sampled instances (494 samples each), ensuring uniform representation and simplifying model training. For the non-IID setting, we employ the Dirichlet distribution to simulate client-level data heterogeneity. In this work, the Dirichlet distribution is used to model label distribution skew across clients, while the feature distribution remains unchanged. Thus, heterogeneity is introduced through differences in class proportions rather than feature perturbations.

Let α be the concentration parameter that controls the skewness of the distribution. Lower α values (e.g., $\alpha = 0.01$) result in highly imbalanced data distributions, while larger values (e.g., $\alpha = 1.0$) yield more IID-like distributions. The Dirichlet distribution is defined as:

$$p(x_1, \ldots, x_k; \alpha_1, \ldots, \alpha_k) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{i=1}^{k} x_i^{\alpha_i - 1}, \tag{1}$$

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where, $x_i \geq 0$ and $\sum_{i=1}^{k} x_i = 1$ represent proportions, α_i controls concentration per class, and $B(\boldsymbol{\alpha}) = \frac{\prod_{i=1}^{k} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{k} \alpha_i\right)}$ is the multivariate Beta function.

To construct the non-IID client partitions, we follow a class-wise Dirichlet allocation strategy commonly used in federated learning. Specifically, for each class c, all samples corresponding to c are first collected. A Dirichlet distribution with parameter α is then sampled across the 20 clients to generate a probability vector representing the proportion of class-c samples assigned to each client. Based on these sampled proportions, the class samples are divided and distributed among the clients. This process is repeated for all classes, and the final client datasets are obtained by aggregating the assigned samples from all classes. We simulate three non-IID scenarios with $\alpha = 1.0$ (low heterogeneity), $\alpha = 0.1$ (moderate), and $\alpha = 0.01$ (high). These distributions are applied under both full and partial client participation settings to reflect real-world variability in data availability and client engagement. This partitioning protocol follows widely adopted FL benchmarking practices to ensure controlled heterogeneity and reproducibility.

3.4. FedAvg Method

FL enables decentralized training while preserving user privacy. We adopt the FedAvg algorithm as the core aggregation mechanism [45]. Each client trains a local model using its private data and sends only the updated model parameters to a central aggregator. Local updates at client k are computed as:

$$w_k^{t+1} = w_k^t - \eta \nabla L_k(w_k^t), \tag{2}$$

where w_k^t is the model at round t, η is the learning rate, and $\nabla L_k(w_k^t)$ is the gradient of the local loss. The global model is updated via weighted averaging:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1}, \tag{3}$$

where n_k is the number of data points at client k and $n = \sum_{k=1}^{K} n_k$ is the total dataset size. The updated model is then broadcast to all clients for the next training round.

3.5. DL Models and Optimizers

We evaluate two deep learning models: Simple 2-Layer Neural Network (2NN) and Tabular Transformer (TabTF). The 2NN comprises an input layer (20 features), a hidden layer with ReLU activation, and an output layer with Softmax activation for binary classification. It offers simplicity and fast training, especially useful in resource-constrained edge devices. On the other hand, TabTF applies self-attention mechanisms to tabular data [46, 47]. Input features are embedded, processed through multi-head self-attention encoder layers, and passed to a feedforward network for classification:

$$\hat{y} = \text{softmax}(W_{\text{out}} \cdot z_{\text{trans}} + b_{\text{out}}), \tag{4}$$

where z_{trans} is the encoder output, and W_{out} and b_{out} are learnable weights and biases.

In our implementation, the TabTF consists of 2 transformer encoder layers with 4 attention heads. The input features are first projected into a 128-dimensional embedding space using a linear embedding layer. Each encoder layer uses a feedforward dimension of 128 with a dropout rate of 0.1 for regularization. TabTF effectively captures feature dependencies in structured data using self-attention mechanisms, making it highly suitable for detecting fraudulent blockchain transactions within FL frameworks.

Unlike sequence transformers, positional encoding is not used because tabular features are treated as independent attributes rather than ordered tokens. The transformer, therefore, operates on embedded feature representations rather than sequential feature positions. The model is governed by four key parameters: the number of input features, the hidden-layer dimensions, the embedding strategy, and dropout-based regularization. Its capacity to model complex, non-linear interactions among features allows it to outperform traditional models on datasets with intricate relationships. Moreover, the TabTF's architectural flexibility makes it particularly valuable in scenarios where conventional models struggle to capture nuanced feature correlations. The architecture is kept lightweight to ensure compatibility with federated environments where client computational resources may be limited.

We compare three optimizers, SGD that performs updates based on gradients of the loss function [48]. Adam combines momentum and adaptive learning rates via first and second moment estimates [49]. Adagrad adjusts learning rates individually per parameter based on accumulated gradient history [50].

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3.6. XAI Techniques

To ensure transparency and trust in predictions, we use two popular XAI techniques: Shapash and LIME. Shapash computes feature contributions using SHAP, which attributes prediction scores to input features based on cooperative game theory. The Shapley value for feature i is:

$$\phi_i = \sum_{S \subseteq N \setminus i} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup i) - f(S)], \tag{5}$$

where S is a subset of features, and $f(S)$ denotes the model output for that subset. Shapash provides both global feature importance and local (instance-level) explanations. On the other hand, LIME constructs a local surrogate model (typically linear regression) around a prediction point x:

$$\mathcal{L}(f, g, \pi_x) = \sum_{z \in Z} \pi_x(z)[f(z) - g(z)]^2, \tag{6}$$

where Z is the set of perturbed samples and $\pi_x(z)$ measures proximity to x. LIME identifies the most influential features contributing to a particular prediction.

In this work, SHAP and LIME explanations are generated only for the final global federated model after the completion of training. We do not generate explanations for individual client models or during intermediate CRs. This design ensures that the explanations reflect the final learned decision behavior of the federated system rather than transient local updates. Furthermore, the explainability analysis is performed centrally after model aggregation and therefore does not introduce additional communication overhead into the FL process. Since SHAP and LIME are applied only during the evaluation phase on a subset of test samples, the computational overhead remains limited and does not affect the training efficiency of the FL framework. From a computational perspective, the cost of SHAP depends on the number of evaluated samples and features, while LIME depends on the number of locally perturbed samples used for surrogate model fitting. Since explanations are generated only once after training, this cost is negligible compared to the cumulative cost of federated optimization involving multiple CRs and local training epochs. This design choice follows common practice in FL-XAI studies, where post-hoc explainability is preferred to avoid additional client-side computation and communication overhead during training.

3.7. Performance Evaluation Metrics

To assess the effectiveness of the model in detecting fraudulent blockchain transactions, we adopt four standard metrics. Accuracy measures the ratio of correctly classified instances, as in Eq. 7. Precision quantifies how many predicted frauds are actual frauds, as in Eq. 8. Recall reflects the ability to detect all fraudulent instances as in Eq. 9. The F1 score balances precision and recall as in Eq. 10. These metrics ensure a robust assessment, particularly in imbalanced datasets such as fraud detection, where positive (fraudulent) instances are rare.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{8}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{9}$$

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{10}$$

4. Results and Discussion

This section presents a comprehensive evaluation of the proposed federated fraud detection framework across IID and non-IID data distributions. Performance is assessed using standard classification metrics and is summarized in Tables 1–4. We compare the performance of two models, namely 2NN and TabTF, trained with three optimizers (SGD, Adam, and AdaGrad) under both full and partial client participation. We also incorporate visualizations of accuracy and loss across CRs and provide detailed global and local explainability using SHAP, Shapash, and LIME.

4.1. Experimental Setup

4.1.1. Hyperparameter Setting

For the partial participation setting, we adopt a stochastic client selection strategy where 50% of the total clients are randomly selected at each CR. This setup follows standard FL practice to simulate realistic scenarios where clients may be unavailable due to communication constraints, energy limitations, or intermittent connectivity. The participating clients are uniformly sampled without replacement at each round to ensure fair participation opportunities across training. Partial participation reduces the number of updates contributing to each aggregation step, which may slightly slow convergence compared to full participation but improves scalability and reflects practical FL deployments. Unless otherwise specified, all experiments with partial participation use a participation ratio of 0.5 across all datasets and models.

To ensure fair comparison across models and optimizers, we use a consistent set of hyperparameters for all experiments. Specifically, we use a batch size of 64, a learning rate of 0.01, and 2 local training epochs per CR. For the IID setting, models are trained for 200 CRs, while for the more challenging non-IID scenarios, we use 500 CRs to ensure stable convergence under heterogeneous data distributions. Hyperparameters were kept fixed across all model–optimizer combinations rather than individually tuned to ensure unbiased performance comparison.

4.1.2. Implementation Details

All experiments were conducted using Python 3.9 and PyTorch 2.0. Key libraries included Scikit-learn, NumPy, Matplotlib, and Pandas. Experiments were run on a Lenovo Legion Pro 7 (Model: MT 82WR BU Idea FM), equipped with an Intel Core i9-13900HX processor, NVIDIA GeForce RTX 4070 (8GB GDDR6), and 32GB RAM.

4.2. IID Performance Analysis

We presents a comparative performance analysis of the 2NN and TabTF models under IID data distributions, utilizing SGD, Adam, and AdaGrad optimizers in Table 1. To ensure statistical robustness, results are reported as mean $\pm$ standard deviation (*std*), the *std* values reflect experimental variance across multiple (5 times) runs.

For the 2NN model under partial client participation, the Adam optimizer demonstrates superior predictive capability, achieving a peak accuracy of 0.8878 ± 0.0189 and an F1 score of 0.9913 ± 0.0162. Notably, Adam's low standard deviation across metrics suggests high model stability, despite a significant computational cost (356.55 ± 5.51 s). In contrast, while SGD offers the highest efficiency (136.13 ± 2.00 s), it suffers from increased performance volatility and a marked reduction in mean accuracy (0.8878 ± 0.0189). AdaGrad provides a robust middle ground, maintaining a competitive accuracy of 0.9893 ± 0.0281 with moderate training overhead. These trends remain consistent under full client participation, where Adam maintains its lead in precision, although the overlapping confidence intervals between AdaGrad and Adam suggest that both are viable for high-accuracy requirements.

The TabTF model consistently outperforms 2NN, highlighting its architectural advantage in capturing complex feature dependencies. Under partial participation, TabTF with Adam reaches a mean accuracy of 0.9904 ± 0.0135. Interestingly, SGD paired with TabTF exhibits remarkable resilience, achieving a high accuracy of 0.9888 ± 0.0279 while being over 2.5 times faster than Adam. This suggests that for Transformer-based architectures in FL, SGD may offer a more favorable efficiency-to-stability ratio.

Under full client participation, TabTF achieves its absolute peak performance with Adam (Acc: 0.9924 ± 0.0296), though the higher variance in training time (791.88 ± 12.31 s) indicates sensitivity to communication overhead. In comparison, the reduced standard deviation in SGD's full participation performance (0.9898 ± 0.0339) reinforces its utility as a reliable alternative for resource-constrained environments. Overall, while TabTF provides the highest predictive ceiling, the inclusion of standard deviations reveals that the 2NN model with AdaGrad offers more predictable and consistent convergence across varying participation levels.

4.3. Non-IID Performance Analysis

We evaluate the performance of the 2NN and TabTF models under varying levels of non-IID data distributions, defined by Dirichlet concentration parameters $\alpha = 1.0$, 0.1, and 0.01. These experiments are conducted using both partial and full client participation with three optimizers (SGD, Adam, and AdaGrad). The results are summarized in Tables 2–4.

4.3.1. Performance under Low Heterogeneity ($\alpha = 1.0$)

We illustrate model performance under low data heterogeneity ($\alpha = 1.0$) in Table 2. In this non-IID setting, the 2NN model paired with the SGD optimizer demonstrates clear superiority in stability and reliability. Specifically, 2NN-SGD

Table 1
Performance of the models under partial and full client participation for IID data distributions with std values over five runs.

Participation	Model	Optimizer	Accuracy	Precision	Recall	F1-Score	Time (Sec)
Partial Client Participation (50% client)	2NN	SGD	0.8878 ± 0.0189	0.8967 ± 0.0345	0.8988 ± 0.0287	0.8784 ± 0.0246	136.13 ± 2.00
		Adam	0.9914 ± 0.0105	0.9914 ± 0.0388	0.9914 ± 0.0347	0.9913 ± 0.0162	356.55 ± 5.51
		AdaGrad	0.9893 ± 0.0281	0.9893 ± 0.0140	0.9893 ± 0.0186	0.9893 ± 0.0208	288.36 ± 6.83
	TabTF	SGD	0.9888 ± 0.0279	0.9889 ± 0.0149	0.9888 ± 0.0118	0.9887 ± 0.0380	280.56 ± 10.93
		Adam	0.9904 ± 0.0135	0.9904 ± 0.0246	0.9904 ± 0.0109	0.9903 ± 0.0369	708.12 ± 12.58
		AdaGrad	0.9761 ± 0.0382	0.9761 ± 0.0325	0.9761 ± 0.0373	0.9758 ± 0.0360	568.25 ± 15.88
Full Client Participation (100% client)	2NN	SGD	0.8878 ± 0.0130	0.8936 ± 0.0105	0.8878 ± 0.0319	0.8747 ± 0.0245	140.20 ± 4.38
		Adam	0.9909 ± 0.0154	0.9909 ± 0.0190	0.9909 ± 0.0255	0.9908 ± 0.0227	366.66 ± 6.87
		AdaGrad	0.9883 ± 0.0332	0.9883 ± 0.0158	0.9883 ± 0.0251	0.9882 ± 0.0274	248.23 ± 2.83
	TabTF	SGD	0.9898 ± 0.0339	0.9899 ± 0.0189	0.9898 ± 0.0128	0.9897 ± 0.0302	238.25 ± 5.53
		Adam	0.9924 ± 0.0296	0.9924 ± 0.0192	0.9924 ± 0.0254	0.9923 ± 0.0262	791.88 ± 12.31
		AdaGrad	0.9782 ± 0.0368	0.9782 ± 0.0124	0.9782 ± 0.0155	0.9779 ± 0.0111	748.17 ± 14.78

Table 2
Performance of the models under non-IID ($\alpha = 1.0$) data distributions with std values over five runs.

Participation	Model	Optimizer	Accuracy	Precision	Recall	F1-Score	Time (Sec)
Partial Client Participation (50% client)	2NN	SGD	0.9858 ± 0.0051	0.9857 ± 0.0048	0.9858 ± 0.0062	0.9857 ± 0.0055	188.89 ± 1.25
		Adam	0.9807 ± 0.0342	0.9807 ± 0.0211	0.9807 ± 0.0415	0.9807 ± 0.0198	399.17 ± 14.22
		AdaGrad	0.9867 ± 0.0285	0.9868 ± 0.0314	0.9868 ± 0.0441	0.9868 ± 0.0276	246.39 ± 8.11
	TabTF	SGD	0.9898 ± 0.0391	0.9898 ± 0.0422	0.9898 ± 0.0185	0.9898 ± 0.0221	1185.83 ± 42.55
		Adam	0.7831 ± 0.0251	0.6133 ± 0.0188	0.7831 ± 0.0362	0.6879 ± 0.0211	1917.24 ± 75.30
		AdaGrad	0.9705 ± 0.0410	0.9703 ± 0.0333	0.9705 ± 0.0456	0.9704 ± 0.0289	1473.75 ± 55.60
Full Client Participation (100% client)	2NN	SGD	0.9837 ± 0.0042	0.9837 ± 0.0055	0.9837 ± 0.0049	0.9837 ± 0.0051	188.98 ± 1.12
		Adam	0.9868 ± 0.0221	0.9868 ± 0.0354	0.9868 ± 0.0412	0.9867 ± 0.0331	249.67 ± 9.44
		AdaGrad	0.9848 ± 0.0311	0.9847 ± 0.0244	0.9848 ± 0.0385	0.9847 ± 0.0298	233.73 ± 7.55
	TabTF	SGD	0.9878 ± 0.0425	0.9878 ± 0.0391	0.9878 ± 0.0155	0.9877 ± 0.0288	1368.73 ± 52.11
		Adam	0.7974 ± 0.0355	0.8117 ± 0.0288	0.7974 ± 0.0311	0.7239 ± 0.0195	1961.11 ± 88.23
		AdaGrad	0.9731 ± 0.0299	0.9729 ± 0.0401	0.9731 ± 0.0322	0.9728 ± 0.0385	1788.07 ± 65.40

achieves a high mean accuracy of 0.9858 ± 0.0051 while maintaining the lowest variance across all metrics (SD $< 0.6\%$), indicating highly predictable convergence despite skewed local data. Comparatively, while the TabTF model reaches slightly higher mean accuracy with SGD (0.9898 ± 0.0391), it exhibits nearly eight times the performance volatility and requires six times more training time (1185.83 ± 42.55 s). The instability of complex architectures is further evidenced by TabTF-Adam, which suffers a significant performance drop (0.7831 ± 0.0251) under partial participation. Overall, 2NN-SGD emerges as the most robust configuration, offering an optimal trade-off between high-confidence predictive output and computational efficiency in decentralized environments.

4.3.2. Performance under Moderate Heterogeneity ($\alpha = 0.1$)

The details of the performance under moderate data heterogeneity ($\alpha = 0.1$) is in Table 3. In this challenging regime, the 2NN model paired with SGD exhibits remarkable structural robustness, maintaining a high accuracy of 0.9842 ± 0.0044 under partial participation with the lowest observed variance across all test cases (SD $< 0.5\%$). This suggests that simple architectures are less susceptible to the gradient explosions and weight oscillations typically triggered by moderate client drift in adaptive optimizers. While Adam performs comparably under full participation (0.9837 ± 0.0492), it shows increased performance volatility under partial participation (0.9792 ± 0.0587). For the TabTF model, although AdaGrad produces the best results under full participation (Acc: 0.9746 ± 0.0294), it suffers from significant latency and resource consumption. Furthermore, TabTF-Adam shows a critical vulnerability to data skew under partial participation, with accuracy dropping to 0.6521 ± 0.0456. Ultimately, the unrivaled efficiency-to-stability ratio of 2NN-SGD makes it the most reliable configuration for resource-constrained FL environments facing moderate statistical heterogeneity.

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Table 3
Performance of the models under non-IID ($\alpha = 0.1$) data distributions with std values over five runs.

Participation	Model	Optimizer	Accuracy	Precision	Recall	F1-Score	Time (Sec)
Partial Client Participation (50% client)	2NN	SGD	0.9842 ± 0.0044	0.9843 ± 0.0039	0.9843 ± 0.0051	0.9841 ± 0.0047	246.93 ± 1.48
		Adam	0.9792 ± 0.0587	0.9793 ± 0.0411	0.9792 ± 0.0632	0.9789 ± 0.0489	283.94 ± 17.04
		AdaGrad	0.9766 ± 0.0391	0.9766 ± 0.0512	0.9766 ± 0.0288	0.9766 ± 0.0614	268.39 ± 13.42
	TabTF	SGD	0.9669 ± 0.0541	0.9667 ± 0.0298	0.9669 ± 0.0483	0.9666 ± 0.0312	1458.14 ± 87.49
		Adam	0.6521 ± 0.0456	0.6757 ± 0.0338	0.6521 ± 0.0261	0.6629 ± 0.0398	2124.93 ± 127.49
		AdaGrad	0.8187 ± 0.0522	0.8405 ± 0.0411	0.8187 ± 0.0394	0.7658 ± 0.0481	1741.48 ± 104.49
Full Client Participation (100% client)	2NN	SGD	0.9787 ± 0.0038	0.9790 ± 0.0041	0.9787 ± 0.0035	0.9788 ± 0.0049	193.41 ± 1.16
		Adam	0.9837 ± 0.0492	0.9837 ± 0.0295	0.9837 ± 0.0512	0.9836 ± 0.0311	253.47 ± 15.21
		AdaGrad	0.9568 ± 0.0612	0.9610 ± 0.0488	0.9568 ± 0.0511	0.9578 ± 0.0384	275.17 ± 16.51
	TabTF	SGD	0.9690 ± 0.0388	0.9691 ± 0.0482	0.9690 ± 0.0511	0.9685 ± 0.0294	1504.22 ± 90.25
		Adam	0.9700 ± 0.0512	0.9698 ± 0.0384	0.9700 ± 0.0295	0.9697 ± 0.0411	2157.97 ± 129.48
		AdaGrad	0.9746 ± 0.0294	0.9749 ± 0.0511	0.9746 ± 0.0388	0.9741 ± 0.0212	1886.69 ± 113.20

Table 4
Performance of the models under non-IID ($\alpha = 0.01$) data distributions with std values over five runs.

Participation	Model	Optimizer	Accuracy	Precision	Recall	F1-Score	Time (Sec)
Partial Client Participation (50% client)	2NN	SGD	0.9624 ± 0.0048	0.9663 ± 0.0038	0.9624 ± 0.0051	0.9633 ± 0.0042	200.13 ± 1.20
		Adam	0.8547 ± 0.0855	0.9001 ± 0.0721	0.8547 ± 0.0812	0.8645 ± 0.0778	321.01 ± 25.68
		AdaGrad	0.7430 ± 0.0743	0.8508 ± 0.0681	0.7430 ± 0.0712	0.7651 ± 0.0765	350.08 ± 28.01
	TabTF	SGD	0.9009 ± 0.0901	0.9066 ± 0.0725	0.9009 ± 0.0811	0.8910 ± 0.0891	1486.11 ± 118.89
		Adam	0.7481 ± 0.0748	0.6801 ± 0.0680	0.7481 ± 0.0711	0.7011 ± 0.0701	2193.86 ± 175.51
		AdaGrad	0.8126 ± 0.0813	0.8822 ± 0.0706	0.8126 ± 0.0731	0.8271 ± 0.0827	1959.58 ± 156.77
Full Client Participation (100% client)	2NN	SGD	0.9711 ± 0.0058	0.9711 ± 0.0041	0.9711 ± 0.0039	0.9711 ± 0.0044	210.02 ± 1.26
		Adam	0.9660 ± 0.0866	0.9676 ± 0.0774	0.9659 ± 0.0869	0.9664 ± 0.0773	257.46 ± 20.60
		AdaGrad	0.6521 ± 0.0652	0.8471 ± 0.0847	0.6521 ± 0.0652	0.6810 ± 0.0681	292.72 ± 23.42
	TabTF	SGD	0.9406 ± 0.0941	0.9439 ± 0.0755	0.9406 ± 0.0847	0.9372 ± 0.0937	1365.71 ± 109.26
		Adam	0.7288 ± 0.0729	0.6695 ± 0.0670	0.7288 ± 0.0729	0.6918 ± 0.0692	2473.73 ± 197.90
		AdaGrad	0.9264 ± 0.0926	0.9248 ± 0.0740	0.9264 ± 0.0834	0.9242 ± 0.0924	2073.19 ± 165.86

4.3.3. Performance under High Heterogeneity ($\alpha = 0.01$)

We present the performance under severe statistical heterogeneity ($\alpha = 0.01$) in Table 4. The 2NN-SGD configuration demonstrates unparalleled resilience in this extreme regime, achieving a high accuracy of 0.9711 ± 0.0058 under full participation. While other models experience severe performance volatility with standard deviations reaching 10%, 2NN-SGD maintains an SD below 0.6%, confirming that non-adaptive gradient updates provide a crucial stabilizing effect when local data distributions are nearly disjoint. In contrast, AdaGrad suffers a catastrophic failure in this setting, with 2NN accuracy dropping to 0.6521 ± 0.0652 and TabTF failing to maintain consistency. While TabTF with SGD offers a competitive mean accuracy of 0.9406 ± 0.0941, its high variance and extreme computational overhead (over 1300 seconds) make it less practical than the 2NN-SGD alternative. Furthermore, TabTF-Adam remains highly unstable, dropping to 0.7288 ± 0.0729. These results reinforce that as heterogeneity increases, minimalist architectures and optimizers emerge as the most dependable solution for high-confidence federated deployments.

4.3.4. Summary of Results and Superior Method Analysis

The comparative analysis across the spectrum of data heterogeneity ($\alpha = 1.0$ to 0.01) reveals a significant divergence between theoretical peak performance and practical reliability in decentralized environments. While the TabTF combined with adaptive optimizers like Adam performs well under IID conditions, it suffers from extreme performance volatility and "client drift" as data becomes increasingly skewed. This phenomenon leads to weight oscillations and gradient instability, particularly in partial participation scenarios where local updates pull the global model in conflicting directions. In contrast, the 2NN model paired with the SGD optimizer emerges as the superior configuration due to its unparalleled structural robustness and variance-reduction capabilities. The superiority of 2NN-SGD is defined by its remarkable consistency; it maintained a high mean accuracy above 96% across all test cases while keeping its *std* below 0.6%, even when complex models reached 10% volatility.

Furthermore, 2NN-SGD proved to be 6 to 10 times more computationally efficient than the TabTF alternatives, requiring significantly fewer resources to achieve convergence. By utilizing non-adaptive gradient updates, this

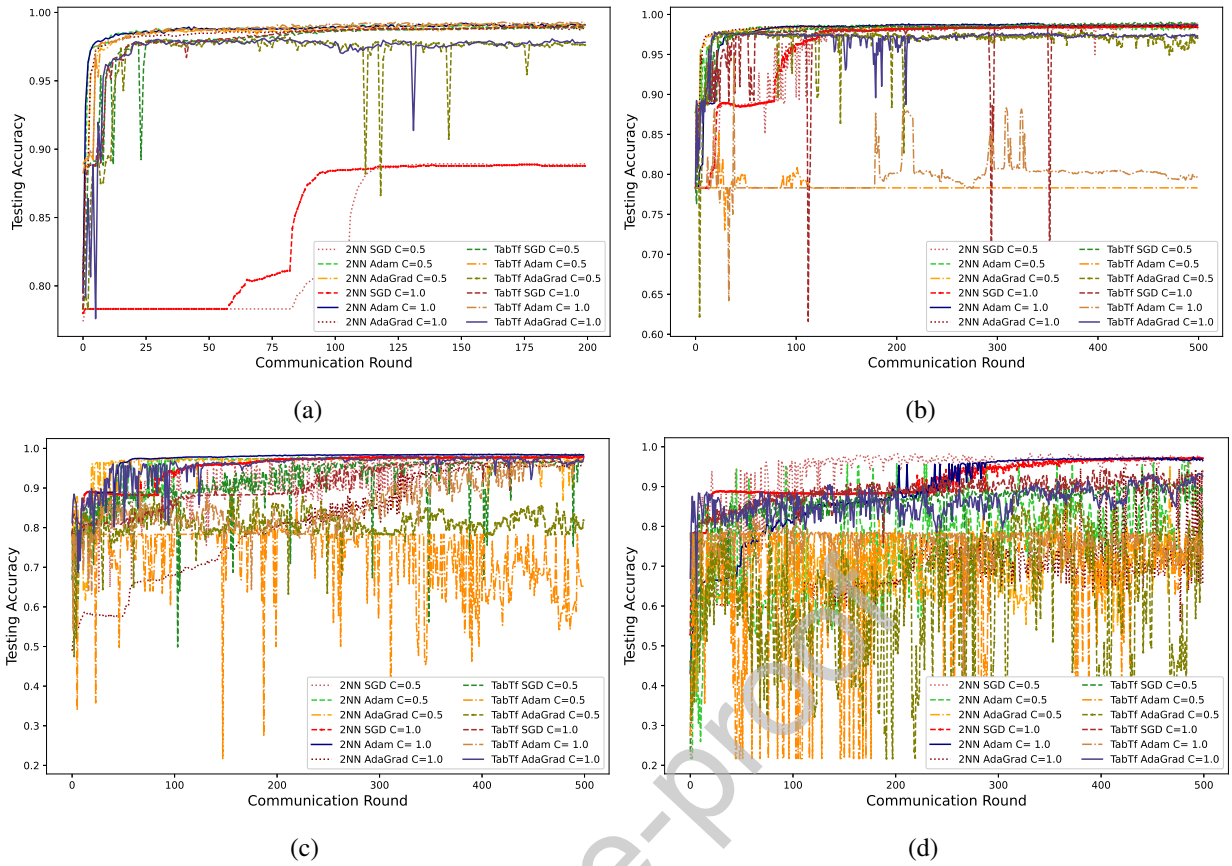


Figure 3: The accuracy of the model vs. CRs for (a) IID data, (b) low-level non-IID data (α=1.0), (c) medium-level non-IID data (α=0.1), (d) high-level non-IID data (α=0.01).

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configuration avoids the overfitting issues common in adaptive algorithms under non-IID regimes. Consequently, 2NN-SGD provides the most predictable and defensible performance profile, making it the optimal choice for high-confidence federated learning systems facing severe statistical heterogeneity.

4.3.5. Performance Visualization

The learning curves of testing accuracy and training loss across communication rounds are present in Figs. 3 and 4, both for IID and non-IID data settings. Fig. 3(a) corresponds to IID, while Fig. 3(b–d) illustrate accuracy trends under $\alpha = 1.0$, 0.1, and 0.01, respectively. Similarly, Fig. 4(a–d) shows the corresponding loss trends for both IID and non-IID. These learning curves provide direct evidence of convergence speed, optimization stability, and training dynamics under varying degrees of statistical heterogeneity. These visualizations reveal the impact of data heterogeneity on convergence stability. For IID and low non-IID scenarios, most models converge smoothly with stable monotonic improvement in accuracy and consistent loss reduction. However, in medium and high non-IID settings, several models, especially TabTF with Adam, exhibit noticeable oscillations in accuracy and slower loss stabilization, reflecting optimization instability caused by heterogeneous client updates. This reflects the difficulty of achieving stable convergence under high client data variability. Despite this, 2NN with SGD consistently shows stable and smooth convergence trajectories across communication rounds and maintains strong accuracy, underscoring its robustness in decentralized, heterogeneous environments. These results demonstrate that increasing data heterogeneity primarily affects convergence stability rather than final performance, highlighting the importance of optimizer and model selection in federated settings.

4.4. Explainability Results

We conduct a comprehensive explainability analysis to elucidate both global and local model behaviors in detecting fraudulent activities within decentralized blockchain transaction data. This interpretability assessment spans global feature importance, individual feature contributions, local instance-based explanations, and cross-instance comparison.

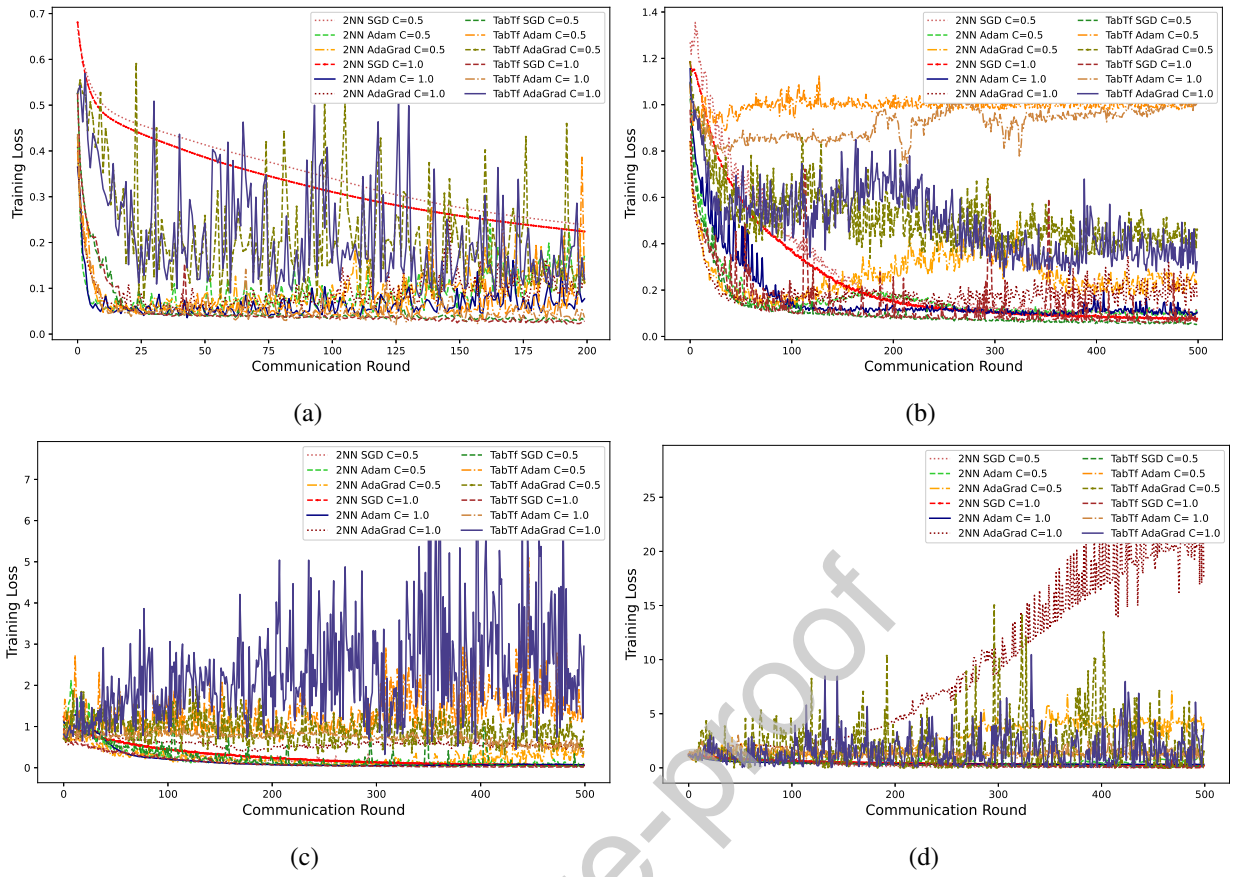


Figure 4: Training loss vs. CRs for (a) IID data, (b) low-level non-IID data (α=1.0), (c) medium-level non-IID data (α=0.1), (d) high-level non-IID data (α=0.01).

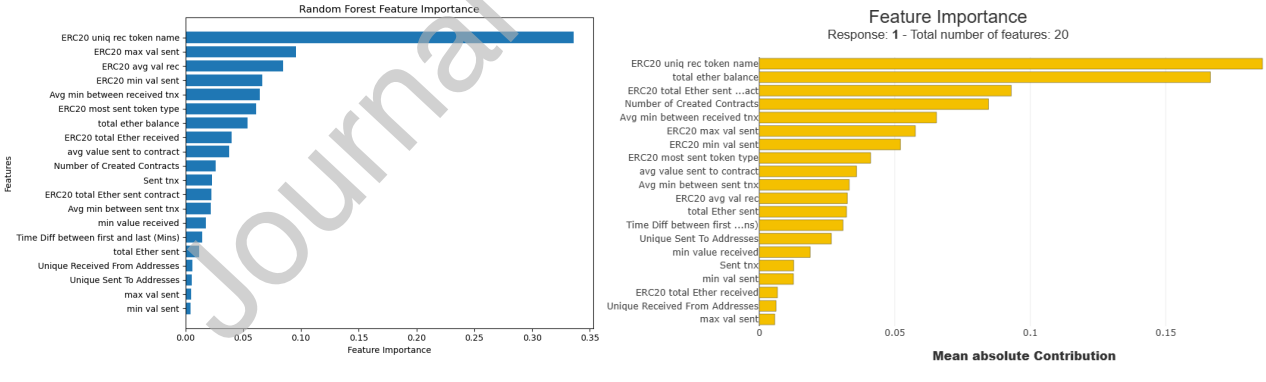


Figure 5: Global explainability (a) Feature importance using RF, (b) Feature importance using Shapash.

The combination of high-level and granular insights enables a deeper understanding of the decision-making process of black-box ML models.

4.4.1. Global Explainability

We illustrate the global feature importance analysis using RF and Shapash, respectively, in Figs. 5 (a) and 5 (b). These techniques offer complementary perspectives on the relative influence of each feature. RF relies on impurity reduction via the Gini index, whereas Shapash utilizes SHAP values derived from game theory to explain the marginal contribution of each feature to model predictions. Across both methods, the feature ERC20 uniq rec token

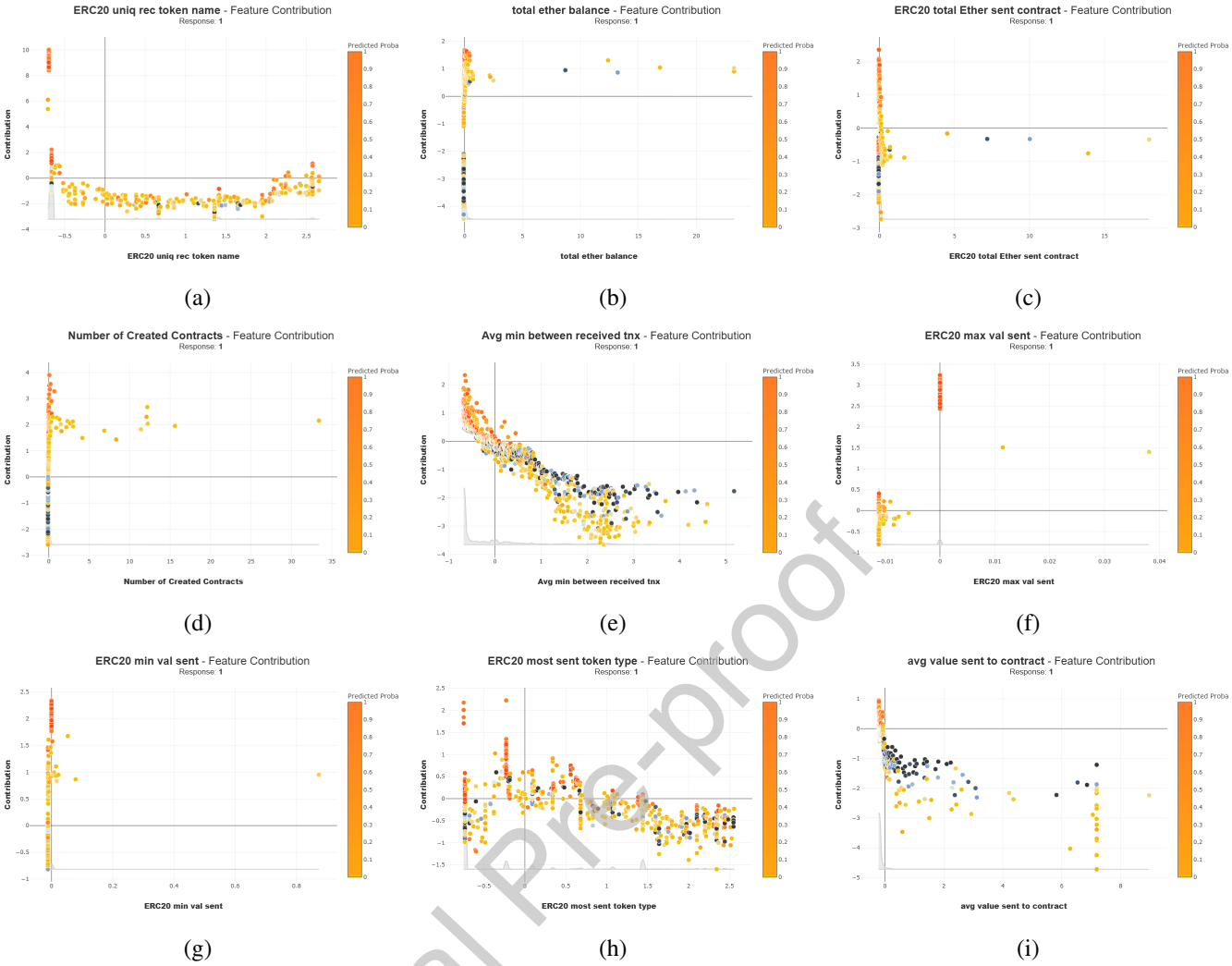


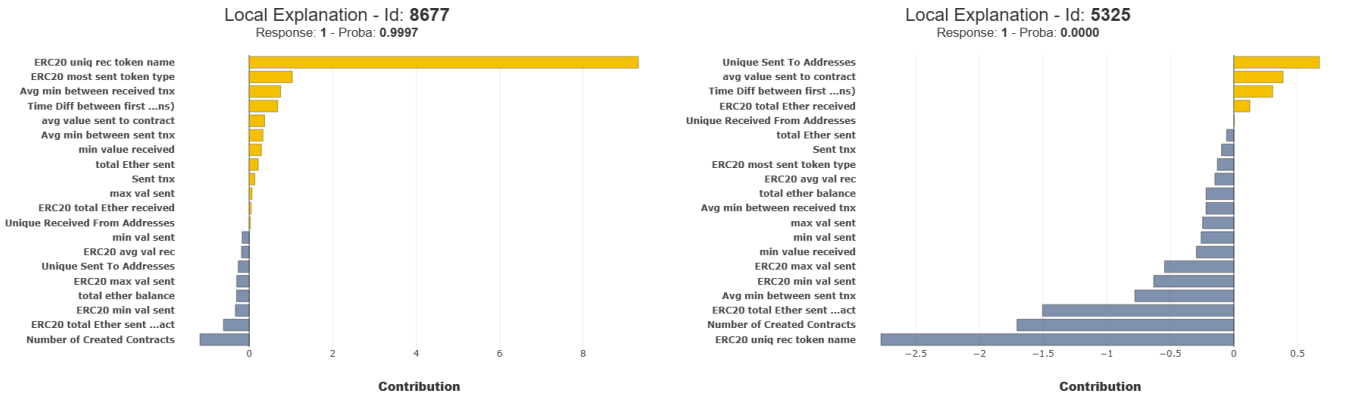
Figure 6: Individual feature importance to show local explanation using Shapash.

name emerges as the most influential predictor of fraud, underscoring the importance of token-specific identifiers in characterizing anomalous behavior. Additionally, ERC20 max val sent, ERC20 avg val rec, and Avg min between received transactions consistently rank highly, highlighting the relevance of transaction values and temporal patterns in identifying fraudulent activity. While RF assigns greater importance to certain features, such as ERC20 max val sent, Shapash emphasizes others, such as total ether balance and ERC20 total Ether sent to the contract. Despite some ranking discrepancies, both techniques converge on a common set of top features, reinforcing their predictive relevance from different interpretability lenses.

4.4.2. Individual Feature Explanations

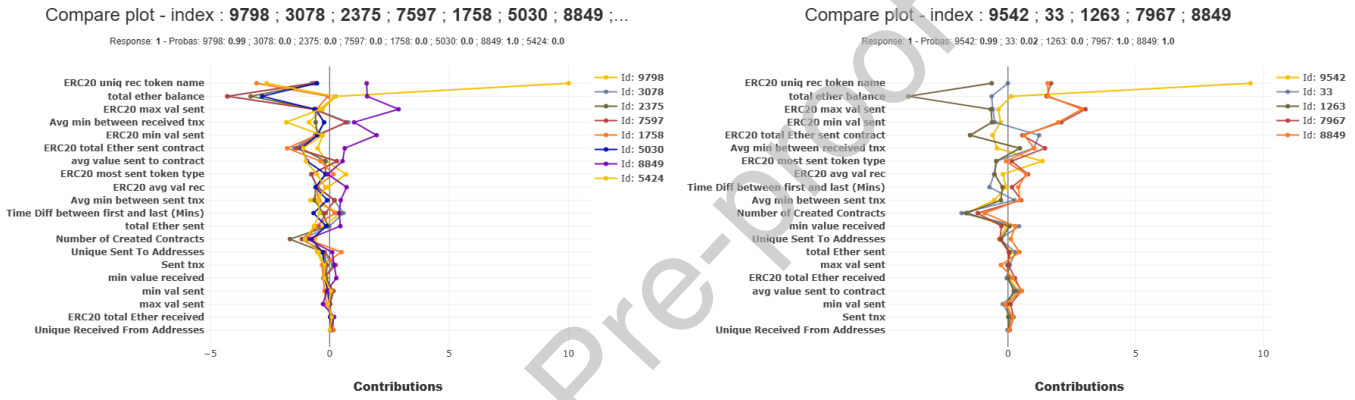
We visualize the influence of the top nine features on the model's predictions using Shapash in Fig. 6 (a–i). The feature ERC20 uniq rec token name displays a strong positive association with fraud probability; higher values of this feature significantly increase the likelihood of fraudulent classification. Similarly, a low total ether balance is often indicative of fraud, although the presence of unusually high balances can also signal risk. The feature ERC20 total Ether sent contract exhibits mixed influence, with both positive and negative contributions depending on its value range. Features such as the Number of created contracts and Avg min between received transactions also demonstrate patterns useful for fraud prediction; frequent transactions and limited contract creation are more common in fraudulent behavior. Notably, larger transaction values (ERC20 max val sent) are strongly associated with fraud, while low-value transactions (ERC20 min val sent) play a more subtle role. Features such as ERC20, the most sent token type, and Avg value sent to the contract exhibit variable contributions, emphasizing the context-dependent nature of fraud indicators.

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(a) (b)

Figure 7: Random instances output probability analysis to show local explanation using Shapash.



(a) (b)

Figure 8: Comparative probability analysis of random five and then instance to show local explanation using Shapash.

4.4.3. Explainability on Random Instances

We present local explanations for high-risk instances in Fig. 7a, while Fig. 7b shows instances with low fraud probability. In high-risk cases, the dominant contributors are ERC20 uniq rec token name, ERC20 most sent token type, and Avg min between received tnx. These features, especially when indicating irregular token behavior and rapid transactions, heavily influence the model toward a fraud prediction. In contrast, low-risk instances are marked by stable values across features like Unique Sent To Addresses and Avg value sent to contract, as well as timing consistency. Interestingly, even features typically associated with fraud may have negative contributions in legitimate cases, reinforcing the importance of contextual interactions in model interpretation.

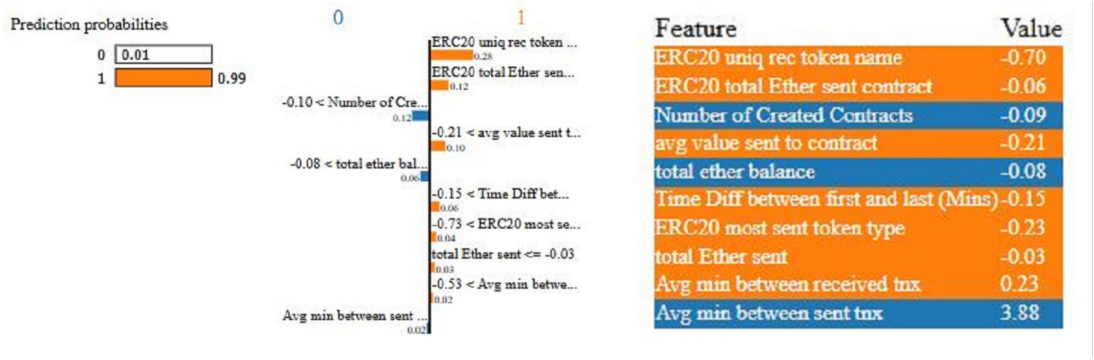
4.4.4. Explainability Across Multiple Instances

We compare feature contributions across ten (Fig. 8a) and five (Fig. 8b) randomly selected instances in Fig. 8. ERC20 uniq rec token name consistently dominates in fraudulent cases but offers negligible or even negative influence in legitimate ones (e.g., IDs 7597 and 5030). Supplementary features like 'total ether balance' and 'Avg min between received tnx' also contribute meaningfully, though their impact varies across cases. This highlights how certain features gain prominence in fraud detection, while others serve supporting roles in specific scenarios.

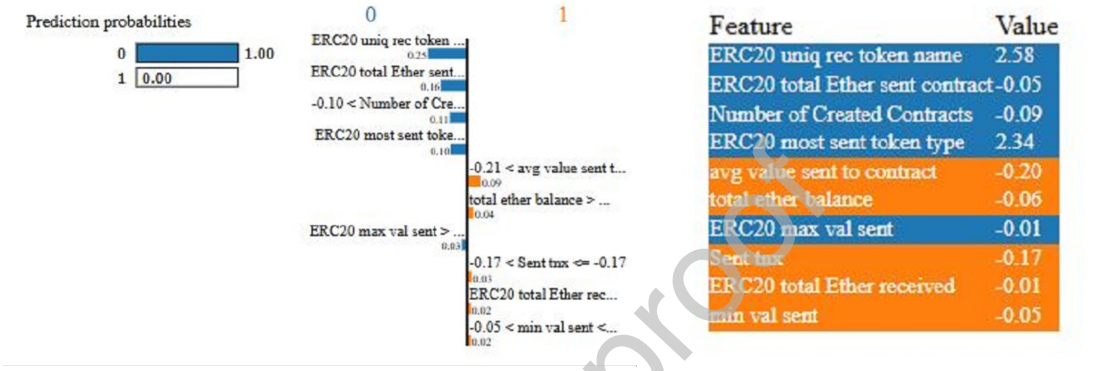
4.4.5. Local Explainability Using LIME

LIME-based explanations in Fig. 9a and 9b further reinforce the above findings. In high-probability fraud predictions (Fig. 9a), ERC20 uniq rec token name and ERC20 most sent token type show strong positive contributions. Conversely, in low-risk predictions (Fig. 9b), these features contribute negatively, supporting the classification of

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(a)



(b)

Figure 9: Random instances output probability analysis to show local explanation using LIME.

legitimate behavior. Features such as the Number of created contracts and Avg value sent to contracts also shift their contributions depending on the instance context. These local explanations from LIME provide case-specific insights that complement Shapash's global interpretations, confirming the model's reliance on token-specific, transactional, and temporal attributes.

4.5. Discussion

This study presents several important insights by integrating performance evaluation with model interpretability. The results highlight optimal combinations of models, optimizers, and participation strategies for various data distributions in a decentralized setting. For IID data, the TabTF model paired with the Adam optimizer under full client participation achieved the highest accuracy and F1 scores, leveraging its advanced feature interaction capabilities. However, this combination is computationally expensive, limiting its deployment in resource-constrained environments. In contrast, 2NN with SGD demonstrated strong performance with significantly faster training times, making it a practical choice for efficiency-focused applications. For non-IID data, model performance is sensitive to the level of data skew. The combination of 2NN and SGD under full participation exhibited superior robustness and efficiency under highly skewed data distributions ($\alpha = 0.01$), maintaining stability and accuracy despite challenging conditions.

XAI techniques support and explain these performance trends. Across both global and local explanations, token-specific features, particularly ERC20 uniq rec token name and ERC20 most sent token type, consistently drive model decisions. For IID data, stable feature distributions contribute to reliable model behavior, especially when using TabTF. In highly non-IID scenarios, timing-related and transactional value features gain importance. The adaptive behavior of 2NN with SGD allows the model to recalibrate and prioritize the most informative features based on the data context, enabling consistent performance. These findings affirm that both performance and interpretability are essential for developing trustworthy fraud detection systems. The alignment between XAI insights and empirical results not only validates model behavior but also enhances user trust. Together, these analyses provide clear recommendations for selecting model architectures, optimizers, and client participation schemes based on data heterogeneity and system constraints.

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5. Conclusion and Future work

This research introduces a privacy-preserving and explainable fraud detection system for blockchain network transactions using FL. The proposed system allows decentralized analysis without sharing raw data, supporting both IID and non-IID scenarios while incorporating XAI to enhance model transparency. Our evaluation demonstrates that TabTF with the Adam optimizer achieves the highest performance under IID distributions, benefiting from full client participation and complex feature interactions. However, this setup demands substantial computational resources. In contrast, the 2NN model with SGD offers a more efficient alternative, delivering competitive performance with significantly lower training time. Under non-IID distributions, the effectiveness of model-optimizer pairs depends heavily on the degree of skew. 2NN with SGD consistently outperformed other combinations under high heterogeneity ($\alpha = 0.01$), providing reliable results with minimal computational overhead.

XAI results validate these findings by revealing the most influential features driving model decisions. Features such as ERC20 uniq rec token name and Avg min between received tnx consistently contribute to fraud prediction across different data distributions and model configurations. These insights ensure that decisions are not only accurate but also interpretable and grounded in data-driven evidence. Looking ahead, future work will focus on improving performance under highly non-IID conditions by developing adaptive optimization strategies and model architectures capable of automatically adjusting to data imbalances. Additionally, further advances in explainability, particularly domain-specific XAI frameworks, will be explored to enhance transparency, user trust, and actionable insights in fraud detection and other blockchain-based applications.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of AI

During the preparation of this work, the authors used ChatGPT to proofread. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

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CRediT authorship contribution statement

Mahmudul Hasan: Conceptualization, Investigation, Data Curation, Methodology, Software, Validation, Verification, Writing – original draft.. **Md Raihanul Islam Tomal:** Data Curation, Methodology, Visualization, Validation, Writing Original Draft. **Md Palash Uddin:** Conceptualization, Investigation, Data Curation, Validation, Verification, Writing – Review & editing, Supervision..

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